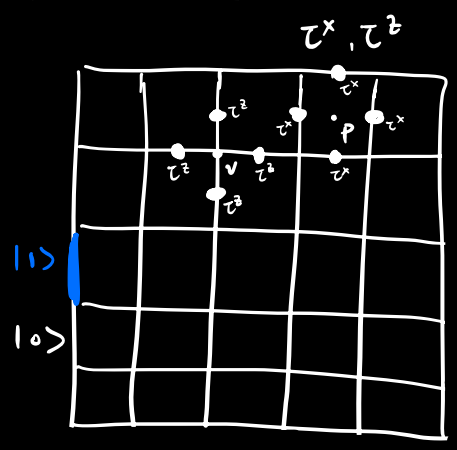
Topological lattice model from Gauging

local symmetry
(constrain)

Toric Code and $\mathbb{Z}_2$ gauge theory



$$H = -\sum_v A_v - \sum_p B_p$$

- $[A_v, A_{v'}] = [B_p, B_{p'}] = 0$

  $[A_v, B_p] = 0$

$\Longrightarrow$ 局部地找 A_v, B_p term 的本征态，即为 H 的本征态；局部地找 ground state，即为 global ground state.

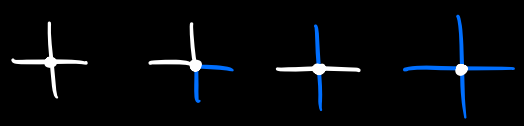
因此，只需局部地分析即可.

- pauli 算符平方 $=1$ $\Longrightarrow$ 本征值 ± 1. 因此基态是本征值为 $+1$ 的态.

能量最低

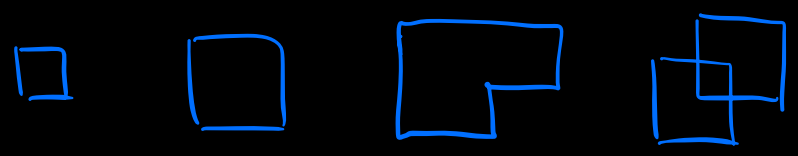
考虑 A_v 的要求：

$$\begin{cases} \tau^z |0\rangle = 1\,|0\rangle & \text{not occupied by a string} \\ \tau^z |1\rangle = -1\,|1\rangle & \text{occupied by a string} \end{cases}$$

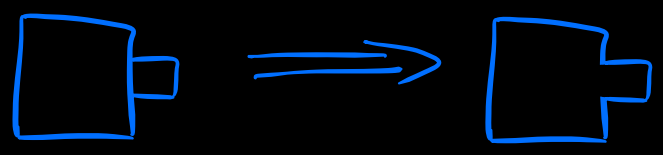
$\Longrightarrow$ we need  and their rotationally symmetric configurations.

一个 vertex 上有偶数个 string $\Longrightarrow$ 一进一出 $\Longrightarrow$ 形成 loops

i.e. $A_v \Longrightarrow$ loops

Rmk  different shapes

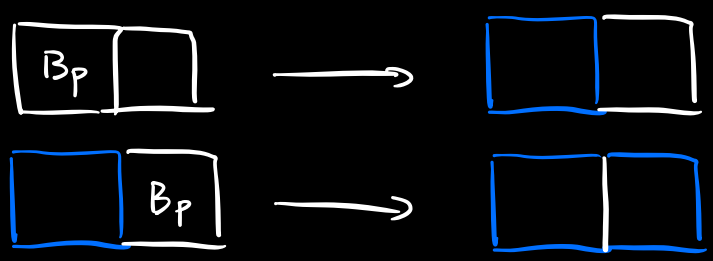
if two loops share boundaries:



($\mathbb{Z}_2$, $-+-=-$)

考虑 B_p 的要求:

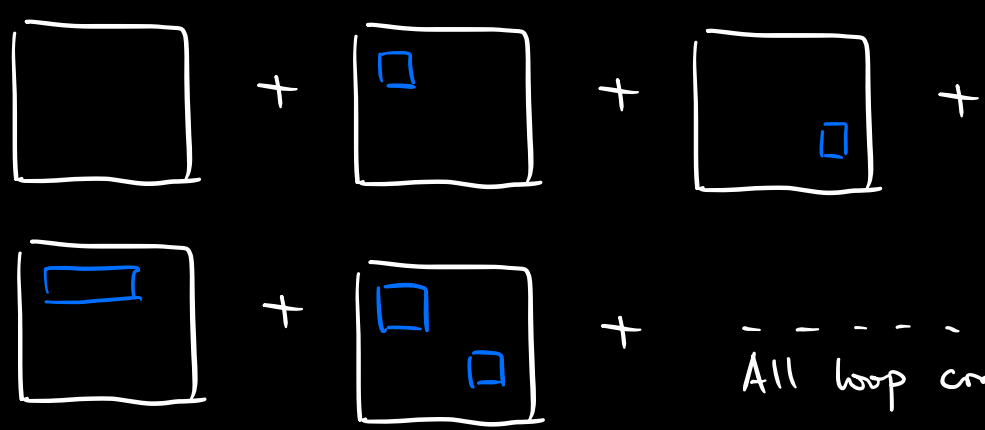
在一个无弦的 lattice 中, B_p 会 create 一个 loop:



(flip twice)

$B_p|\psi\rangle = |\psi\rangle$ (+1 eigenvalue)

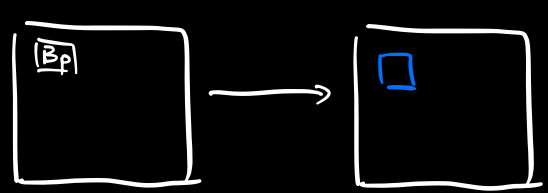
↳ ground state wavefunction

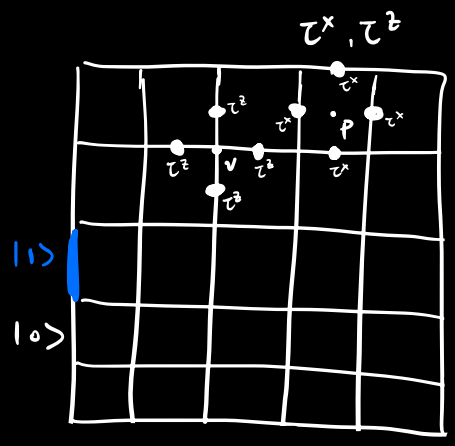
$|\psi\rangle =$ 

All loop configurations

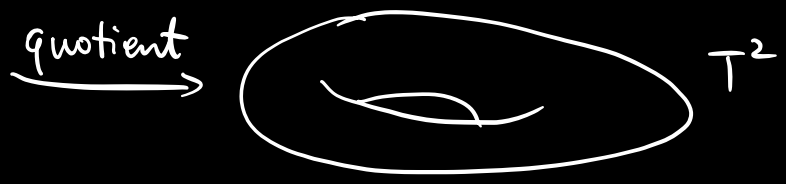
Rmk 在所有 configurations 中, B_p 可以在其中将一个相映射到另一个相中.

e.g.

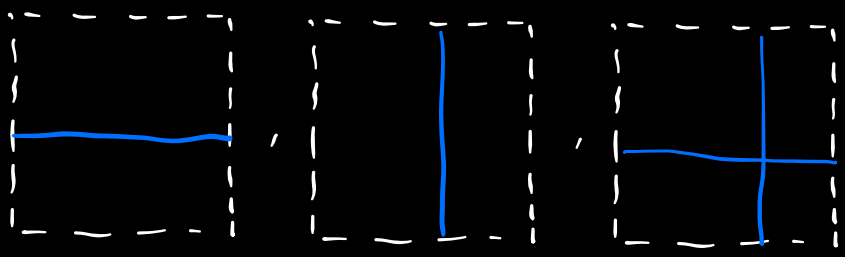




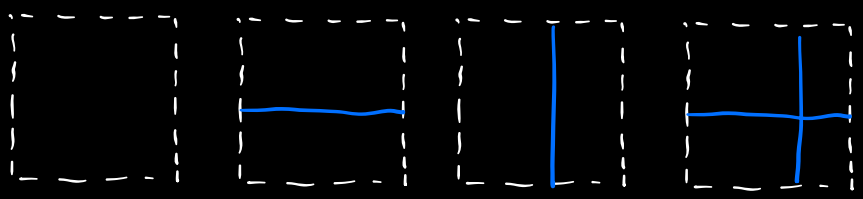
若将此二维系统的四个边界 quotient 掉使其成为一个 Torus，情形就不同了！



等价地，可以说采用 periodic boundary condition (PBC)

此时，  也是 loops.

这种 loop 不能 created by B_p.



从这四种情形出发，我们可以用 B_p 在其中 create 那些可缩的 loops；其态的 Hilbert space 分成四个子空间，即：

在 上，有 4-fold ground state degeneracy.

* 在 Sphere S^2 上，仅有 1-fold，因为球面上任一 loop 均可缩 (Non-degenerate)

<u>Rmk</u> 2 constrains: $\underbrace{\prod_v A_v = \mathbb{1}, \quad \prod_p B_p = \mathbb{1}}$

Dimension of total Hilbert space 4^{N_v}

2 Hamiltonian terms per unit cell $\Rightarrow 4^{N_v}/(4^{N_v}/4) = 4$

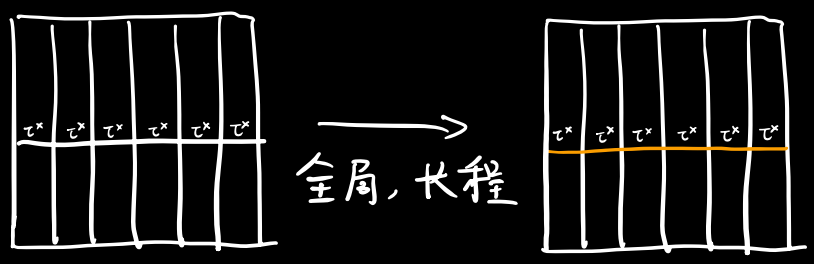
<u>Rmk</u> 4-fold 简并对 $\forall$ local 微扰是 stable 的.

* 为了说明这点，需要考查如何实现不同基态间的映射，(e.g.)

发现这需要 global，long-range 的操作，局部微扰办不到.

Logical operator : 实现不同 ground states 间的 maps.

e.g.



$$\begin{cases} H|\psi_1\rangle = E_{min}|\psi_1\rangle \\ H|\psi_2\rangle = E_{min}|\psi_2\rangle \end{cases}$$

$$\langle\psi_1|\prod_L \tau^x|\psi_2\rangle = 1$$

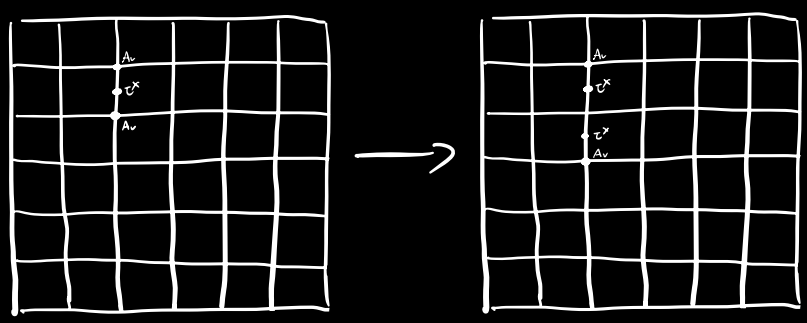
$$H + \varepsilon\sum_i \tau^x$$

$$\langle\psi_1|\tau^x|\psi_2\rangle = 0$$

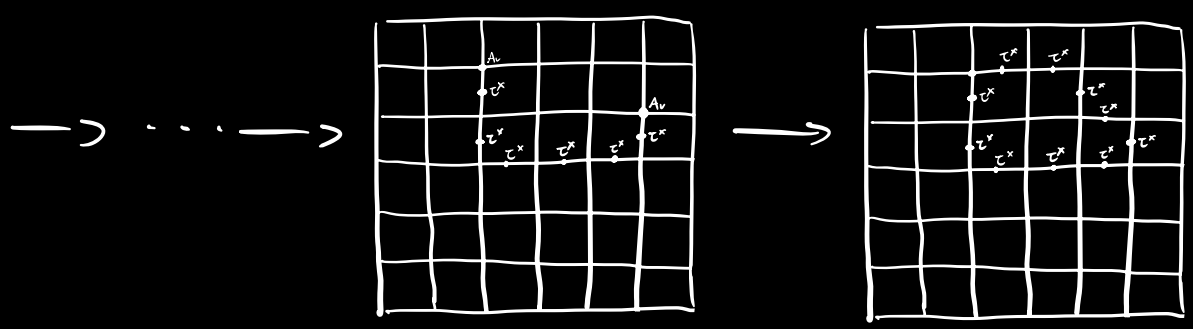
$$\langle\psi_1|\varepsilon^L\prod_L \tau^x|\psi_2\rangle \propto \varepsilon^L$$

Excitation:

τ^x :



τ^x 创造了两个顶点的激发，随后用 τ^x 作用可以将这两个激发分开



若走一个loop，则回到基态.

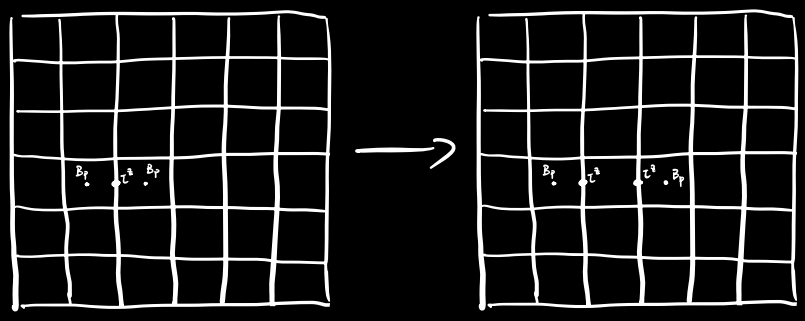
Violation of A_v $\Longrightarrow$ called a charge excitation

$$* \begin{cases} B_p\tau^x|\psi\rangle = \tau^x B_p|\psi\rangle = \tau^x|\psi\rangle \\ A_v\tau^x|\psi\rangle = -\tau^x A_v|\psi\rangle = -\tau^x|\psi\rangle \end{cases}, \quad |\psi\rangle: \text{ground state}$$

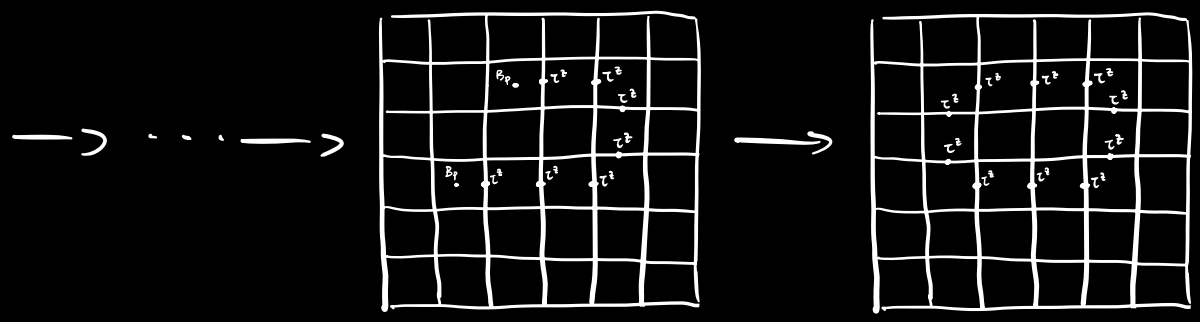
(A_v is excited)

τ^z :

$$\begin{cases} A_v\tau^z|\psi\rangle = \tau^z A_v|\psi\rangle = \tau^z|\psi\rangle \\ B_p\tau^z|\psi\rangle = -\tau^z B_p|\psi\rangle = -\tau^z|\psi\rangle \end{cases}, \quad |\psi\rangle: \text{ground state}$$



τ^z 创造了两个顶点的激发，随后用 τ^z 作用可以将这两个激发分开



若走一个loop，则回到基态

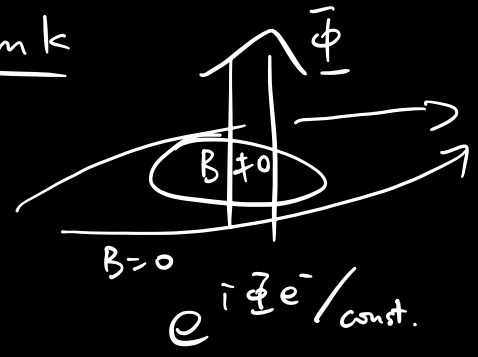
Violation of B_p $\Longrightarrow$ called a flux excitation

* Charge, flux excitation are the two basis excitation of toric code.

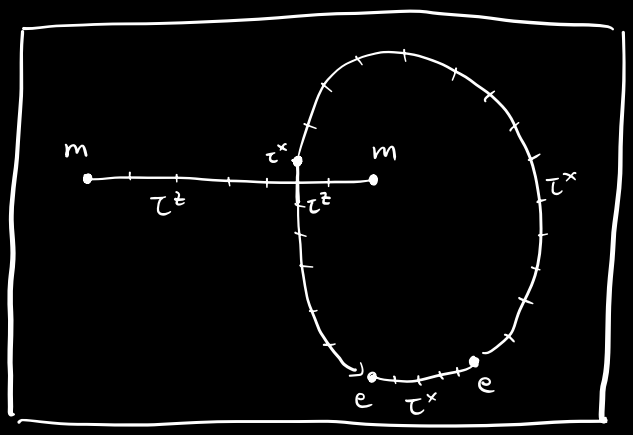
| notation |

e m

Rmk



猜测



The Aharonov-Bohm Effect

转一圈差一个相位 (Berry phase)

果然，由于 τ^x 和 τ^z 反对易，先绕圈后造 m 和先造 m 后绕圈差一个负号。

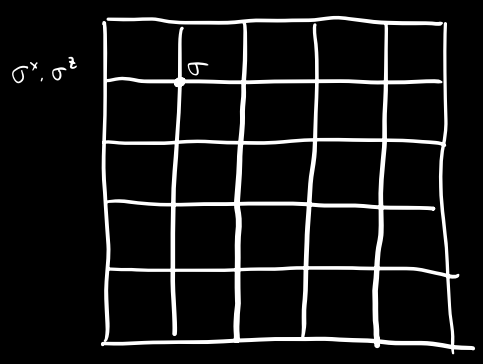
因此这里也有 A-B effect.

$e^{i\pi}$

Rmk

$\Longrightarrow$ e, m are bosons.

Transverse field Ising model



$$H = -J \sum_{\langle i,j \rangle} \sigma_i^x \sigma_j^x - \sum_i \sigma_i^z$$

$$U = \prod \sigma_i^z \qquad HU = UH$$

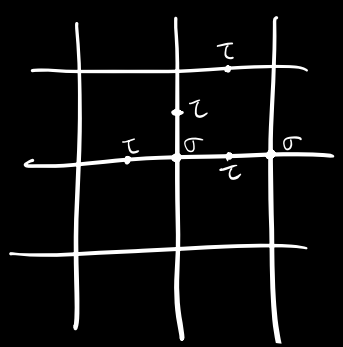
$J << 1$ symmetric phase

$J >> 1$ symmetric breaking phase

Couple TFI model to $\mathbb{Z}_2$ Gauge field

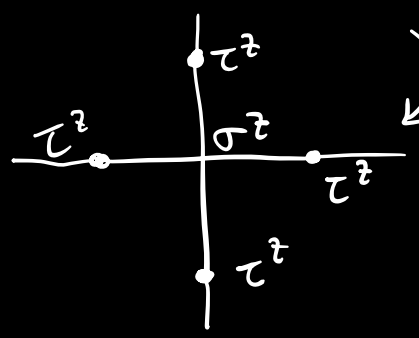
|

local



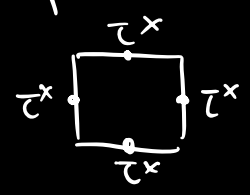
we want to couple them together to get local symmetry.

local symmetry operator $\quad (\mathbb{Z}_2 : \ \cdot^2 = \mathbb{1})$



是这5个家伙的 tensor product

zero-flux condition



$$\rightsquigarrow \quad H_g = -J \sum_{\langle i,j \rangle} \sigma_i^x \tau_{\langle i,j \rangle}^x \sigma_j^x - \sum_i \sigma_i^z - \sum_p B_p$$

↳ 消除简并

$J=0$: $H_g = -\sum_i \sigma_i^z - \sum_p B_p$

$$= -\sum_v A_v - \sum_p B_p$$

local symmetry 的存在，使所有东西心结是它的 eigenstate，因此有

Local Sym $|\sigma^z\rangle = |\sigma^z\rangle$ (eigenfunction)

$= |\tau^z\rangle$ (since $\sigma^{x^2} = \mathbb{1}$)

$\rightsquigarrow$ 正是 A_v

Sym. phase of TFI model with global $\mathbb{Z}_2$ - sym

$\Downarrow$ 与 $\mathbb{Z}_2$ - gauge field 耦合，使得 model becomes locally symmetric.

Toric code with topological order

Summary

1. Dynamics for matter field is the same at low energy.
2. Gauge charge comes from symmetry charge. $\quad e \times e = 1 \longrightarrow$ sym.
   ↳ bosonic $\longrightarrow$ statistic of sym. charge
3. AB phase $\longrightarrow$ determined by sym. group.